MERN Stack Interview Questions and Answers

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1. Introduction to MERN Stack

Q1. What is the MERN Stack?

Answer:
MERN Stack is a full-stack JavaScript technology stack used to build modern web applications. MERN stands for:

Technology	Purpose
MongoDB	NoSQL database
Express.js	Backend web framework
React.js	Frontend library
Node.js	JavaScript runtime environment

The main advantage of MERN is that JavaScript is used across the entire application, from frontend to backend.

Q2. Why is MERN Stack popular?

Answer:
MERN Stack is popular because:

- It uses JavaScript for both frontend and backend.
- React provides fast and interactive UI development.

- Node.js handles asynchronous operations efficiently.
- MongoDB stores flexible JSON-like documents.
- Express.js simplifies backend API development.
- It is widely used for scalable web applications.

Q3. What type of applications can be built using MERN Stack?

Answer:
MERN Stack can be used to build:

- E-commerce websites
- Social media platforms
- Portfolio websites
- Admin dashboards
- Blogging platforms
- Real-time chat applications
- Learning management systems
- Job portals
- SaaS applications

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2. MongoDB Interview Questions

Q4. What is MongoDB?

Answer:
MongoDB is a NoSQL database that stores data in flexible, JSON-like documents called BSON. Unlike relational databases, MongoDB does not store data in tables and rows. Instead, it stores data in collections and documents.

Example document:

```JSON
{
"name": "Dipanshu",
"email": "dipanshu@example.com",
"age": 22
}
```

Q5. What is the difference between SQL and MongoDB?

Answer:

SQL Database	MongoDB
Stores data in tables	Stores data in collections
Uses rows and columns	Uses documents
Has fixed schema	Has flexible schema
Uses SQL queries	Uses MongoDB query syntax
Good for structured data	Good for flexible and scalable data

Q6. What is a collection in MongoDB?

Answer:

A collection in MongoDB is a group of documents. It is similar to a table in SQL databases.

Example:

- users collection
- products collection
- orders collection

Each collection contains multiple documents.

Q7. What is a document in MongoDB?

Answer:

A document is a single record in MongoDB. It stores data in key-value pairs.

Example:

```JSON
{
"_id": "101",
"name": "Rahul",
```

```
"role": "Frontend Developer"
}
```

Q8. What is BSON?

Answer:

BSON stands for Binary JSON. MongoDB stores documents internally in BSON format. BSON is similar to JSON but supports more data types such as Date, ObjectId, and binary data.

Q9. What is ObjectId in MongoDB?

Answer:

ObjectId is a unique identifier automatically generated by MongoDB for each document. It is stored in the `_id` field.

Example:

```JSON
{
"_id": "507f1f77bcf86cd799439011",
"name": "Amit"
}
```

Q10. What are indexes in MongoDB?

Answer:

Indexes improve the speed of search queries in MongoDB. Without indexes, MongoDB scans every document in a collection.

Example:

```JS
db.users.createIndex({ email: 1 })
```

This creates an index on the `email` field.

Q11. What is Mongoose?

Answer:

Mongoose is an Object Data Modeling library for MongoDB and Node.js. It helps define schemas, models, validations, and relationships for MongoDB data.

Example:

```JS
const userSchema = new mongoose.Schema({
name: String,
email: String,
password: String
});
```

Q12. What is a schema in Mongoose?

Answer:

A schema defines the structure of documents in a MongoDB collection. It specifies fields, data types, validations, and default values.

Example:

```JS
const productSchema = new mongoose.Schema({
name: String,
price: Number,
inStock: Boolean
});
```

Q13. What is a model in Mongoose?

Answer:

A model is created from a schema and is used to interact with the database.

Example:

```JS
const Product = mongoose.model("Product", productSchema);
```

Using this model, we can create, read, update, and delete products.

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3. Express.js Interview Questions

Q14. What is Express.js?

Answer:
Express.js is a lightweight backend framework for Node.js. It is used to build APIs and web applications easily.

Example:

```js
const express = require("express");
const app = express();


app.get("/", (req, res) => {
res.send("Hello World");
});

app.listen(5000);
```

Q15. Why do we use Express.js?

Answer:
Express.js is used because it:

- Simplifies route handling
- Supports middleware
- Makes API development easier
- Handles HTTP requests and responses
- Works well with MongoDB and Node.js
- Provides a clean structure for backend applications

Q16. What is routing in Express.js?

Answer:
Routing defines how an application responds to client requests for specific URLs and HTTP methods.

Example:

```js
app.get("/users", (req, res) => {
res.send("All users");
});


app.post("/users", (req, res) => {
res.send("User created");
});
```

Q17. What are middleware functions in Express.js?

Answer:

Middleware functions are functions that execute between the request and response cycle. They can modify the request, response, or stop the request.

Example:

```js
app.use(express.json());
```

Common uses of middleware:

- Authentication
- Logging
- Error handling
- Parsing JSON data
- Handling CORS

Q18. What is `req` and `res` in Express?

Answer:

Object	Meaning
`req`	Request object containing client request data
`res`	Response object used to send response back to client

Example:

```JS
app.get("/profile", (req, res) => {
res.json({ message: "Profile data" });
});
```

Q19. What is CORS?

Answer:

CORS stands for Cross-Origin Resource Sharing. It allows a frontend application running on one domain or port to access backend APIs running on another domain or port.

Example:

`Frontend: http://localhost:3000`

`Backend: http://localhost:5000`

To allow communication, we use CORS middleware:

```JS
const cors = require("cors");
app.use(cors());
```

Q20. What is error-handling middleware in Express?

Answer:

Error-handling middleware is used to handle errors in one central place.

Example:

```JS
app.use((err, req, res, next) => {
res.status(500).json({
message: err.message
});
});
```

It has four parameters: `err`, `req`, `res`, and `next`.

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4. React.js Interview Questions

Q21. What is React.js?

Answer:

React.js is a JavaScript library used to build user interfaces. It is mainly used for building single-page applications where data changes dynamically without refreshing the page.

Q22. What are components in React?

Answer:

Components are reusable building blocks of a React application. Each component represents a part of the UI.

Example:

```jsx
function Header() {
return <h1>Welcome</h1>;
}
```

There are two main types:

- Functional components
- Class components

Modern React mostly uses functional components.

Q23. What is JSX?

Answer:

JSX stands for JavaScript XML. It allows us to write HTML-like syntax inside JavaScript.

Example:

```jsx
const heading = <h1>Hello MERN</h1>;
```

JSX makes UI code easier to read and write.

Q24. What is the Virtual DOM?

Answer:

The Virtual DOM is a lightweight copy of the real DOM. React updates the Virtual DOM first, compares it with the previous version, and then updates only the changed parts in the real DOM. This improves performance.

Q25. What is state in React?

Answer:

State is used to store data that can change over time in a component.

Example:

```jsx
const [count, setCount] = useState(0);
```

When state changes, React re-renders the component.

Q26. What are props in React?

Answer:

Props are used to pass data from one component to another, usually from parent to child.

Example:

```jsx
function Welcome(props) {
return <h1>Hello {props.name}</h1>;
}
```

Props are read-only.

Q27. Difference between state and props?

Answer:

State	Props
Managed inside component	Passed from parent component
Can be changed	Read-only
Used for dynamic data	Used for communication
Controlled by component itself	Controlled by parent

Q28. What is `useState` hook?

Answer:

useState is a React hook used to add state to functional components.

Example:

```jsx
const [name, setName] = useState("Dipanshu");
```

Here:

- `name` is the state variable
- `setName` is the function used to update it

Q29. What is `useEffect` hook?

Answer:

useEffect is used to perform side effects in React components.

Common use cases:

- Fetching API data
- Updating document title
- Setting timers
- Subscribing to events

Example:

```
```

```jsx
useEffect(() => {
console.log("Component mounted");
}, []);
```

Q30. What does the dependency array in `useEffect` do?

Answer:

The dependency array controls when `useEffect` runs.

```jsx
useEffect(() => {
// Runs only once
}, []);
```

```jsx
useEffect(() => {
// Runs when count changes
}, [count]);
```

If no dependency array is provided, `useEffect` runs after every render.

Q31. What is React Router?

Answer:

React Router is a library used for handling routing in React applications. It helps create multiple pages without refreshing the browser.

Example routes:

- `/`
- `/about`
- `/login`
- `/dashboard`

Q32. What is conditional rendering in React?

Answer:

Conditional rendering means displaying UI based on a condition.

Example:

```jsx
{isLoggedIn ? <Dashboard /> : <Login />}
```

If `isLoggedIn` is true, Dashboard is shown. Otherwise, Login is shown.

Q33. What are keys in React lists?

Answer:

Keys help React identify which items have changed, been added, or removed from a list.

Example:

```jsx
users.map(user => <li key={user.id}>{user.name}</li>)
```

Keys should be unique.

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5. Node.js Interview Questions

Q34. What is Node.js?

Answer:

Node.js is a JavaScript runtime environment that allows JavaScript to run outside the browser. It is commonly used for backend development.

Q35. Why is Node.js fast?

Answer:

Node.js is fast because:

- It uses Google Chrome's V8 JavaScript engine.
- It has a non-blocking I/O model.

- It uses an event-driven architecture.
- It can handle multiple requests efficiently.

Q36. What is npm?

Answer:

npm stands for Node Package Manager. It is used to install, manage, and share JavaScript packages.

Example:

```bash
npm install express
```

Q37. What is `package.json`?

Answer:

`package.json` stores project information, dependencies, scripts, and configuration.

Example:

```json
{
"name": "mern-app",
"version": "1.0.0",
"scripts": {
"start": "node server.js"
}
}
```

Q38. What is the difference between dependencies and devDependencies?

Answer:

dependencies	devDependencies
Required in production	Required only during development
Example: express, mongoose	Example: nodemon, eslint

Installed normally	Installed using `--save-dev`

Example:

```bash
npm install express
npm install nodemon --save-dev
```

Q39. What is the event loop in Node.js?

Answer:

The event loop allows Node.js to perform non-blocking operations. It handles asynchronous tasks like file reading, API calls, and database operations without blocking the main thread.

Q40. What is callback hell?

Answer:

Callback hell occurs when multiple nested callbacks make code difficult to read and maintain.

Example:

```js
getUser(id, function(user) {
getOrders(user.id, function(orders) {
getPayment(orders[0].id, function(payment) {
console.log(payment);
});
});
});
```

It can be avoided using Promises or async/await.

Q41. What is async/await?

Answer:

`async/await` is used to write asynchronous JavaScript code in a cleaner way.
Example:

```js
async function getUsers() {
const users = await User.find();
return users;
}
```

It makes asynchronous code look similar to synchronous code.

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6. MERN Backend API Questions

Q42. What is REST API?

Answer:
REST API is an architectural style used to create web services. It allows frontend and backend to communicate using HTTP methods.

Common HTTP methods:

Method	Purpose
GET	Read data
POST	Create data
PUT	Update full data
PATCH	Update partial data
DELETE	Delete data

Q43. How does frontend communicate with backend in MERN?

Answer:
In MERN Stack, React frontend communicates with Express/Node backend using HTTP requests.

Common libraries:

- `fetch`
- `axios`

Example:

```JS
const response = await axios.get("http://localhost:5000/api/users");
```

The backend sends data as JSON.

Q44. What is CRUD?

Answer:

CRUD stands for:

Operation	Meaning	HTTP Method
Create	Add new data	POST
Read	Get data	GET
Update	Modify data	PUT/PATCH
Delete	Remove data	DELETE

Q45. Explain a basic MERN application flow.

Answer:

1. User interacts with the React frontend.
2. React sends an API request to the Express backend.
3. Express receives the request.
4. Backend communicates with MongoDB using Mongoose.
5. MongoDB returns data.
6. Express sends response to React.
7. React updates the UI.

Q46. What is MVC architecture?

Answer:

MVC stands for Model-View-Controller.

Part	Role
Model	Handles database logic
View	Handles UI
Controller	Handles request and response logic

In MERN:

- Model: Mongoose schema/model
- View: React components
- Controller: Express controller functions

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7. Authentication and Security Questions

Q47. What is authentication?

Answer:
Authentication is the process of verifying who a user is.

Example:

- Login using email and password
- OTP verification
- Google login

Q48. What is authorization?

Answer:
Authorization checks what an authenticated user is allowed to access.

Example:

- Normal user can view products
- Admin can add or delete products

Authentication asks: “Who are you?”
Authorization asks: “What are you allowed to do?”

Q49. What is JWT?

Answer:

JWT stands for JSON Web Token. It is used for securely transmitting user information between frontend and backend.

A JWT usually contains:

- Header
- Payload
- Signature

After login, the server generates a token and sends it to the client. The client sends this token with future requests.

Q50. How does JWT authentication work in MERN?

Answer:

1. User enters login details.
2. Backend verifies email and password.
3. Backend creates a JWT token.
4. Token is sent to frontend.
5. Frontend stores token.
6. Frontend sends token in request headers.
7. Backend verifies token before giving access.

Example header:

```
JS
Authorization: Bearer token_here
```

Q51. What is bcrypt?

Answer:

bcrypt is a library used to hash passwords before storing them in the database. It protects user passwords even if the database is compromised.

Example:

```
```

```
JS
const hashedPassword = await bcrypt.hash(password, 10);
```

Q52. Why should passwords not be stored as plain text?

Answer:

Passwords should not be stored as plain text because if the database is hacked, attackers can directly see user passwords. Passwords should always be hashed using libraries like bcrypt.

Q53. What are environment variables?

Answer:

Environment variables store sensitive configuration values outside the code.

Example:

```
ENV
MONGO_URI=mongodb_connection_string
JWT_SECRET=mysecretkey
PORT=5000
```

They are usually stored in a `.env` file.

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8. Advanced MERN Questions

Q54. What is middleware authentication?

Answer:

Authentication middleware checks whether a user is logged in before allowing access to protected routes.

Example:

```
JS
const authMiddleware = (req, res, next) => {
const token = req.headers.authorization;


if (!token) {
return res.status(401).json({ message: "No token provided" });
}
```

```
next();
};
```

Q55. What is protected routing in React?

Answer:

Protected routing prevents unauthorized users from accessing private pages.

Example:

- Logged-in user can access `/dashboard`
- Non-logged-in user is redirected to `/login`

Q56. What is Redux?

Answer:

Redux is a state management library used to manage global application state. It is useful when data needs to be shared across many components.

Example use cases:

- User authentication state
- Cart items
- Theme settings
- Application preferences

Q57. What is Context API?

Answer:

Context API is a built-in React feature used to share data between components without passing props manually at every level.

It is useful for:

- Theme management
- User login state
- Language settings

Q58. Difference between Context API and Redux?

Answer:

Context API	Redux
Built into React	External library
Good for simple state sharing	Good for complex state management
Less boilerplate	More structured
Suitable for small/medium apps	Suitable for large apps

Q59. What is pagination?

Answer:

Pagination is the process of dividing large data into smaller pages.

Example:

Instead of showing 10,000 products at once, we show 10 or 20 products per page.

Benefits:

- Faster loading
- Better user experience
- Reduced server load

Q60. What is server-side validation?

Answer:

Server-side validation means checking user input on the backend before saving it to the database.

Example:

- Email must be valid
- Password must be strong
- Required fields must not be empty

Even if frontend validation exists, backend validation is still necessary for security.

Q61. What is client-side validation?

Answer:

Client-side validation happens in the browser before data is sent to the server.

Example:

```js
if (!email) {
alert("Email is required");
}
```

It improves user experience but should not replace backend validation.

Q62. What is deployment in MERN?

Answer:

Deployment means making the MERN application live on the internet.

Common deployment platforms:

Part	Platform
Frontend	Vercel, Netlify
Backend	Render, Railway, Heroku alternatives
Database	MongoDB Atlas

Q63. What is MongoDB Atlas?

Answer:

MongoDB Atlas is a cloud database service for MongoDB. It allows developers to create, manage, and scale MongoDB databases online.

Q64. What is the role of `.env` file in MERN?

Answer:

The `.env` file stores sensitive data and configuration values.

Example:

```env
PORT=5000
MONGO_URI=your_mongodb_connection_url
JWT_SECRET=your_jwt_secret
```

It helps keep secret keys and database URLs safe.

Q65. What are common security practices in MERN?

Answer:

Common security practices include:

- Hash passwords using bcrypt
- Use JWT securely
- Validate input on backend
- Use environment variables
- Enable CORS properly
- Sanitize user input
- Use HTTPS in production
- Avoid exposing secret keys
- Implement role-based access control
- Use rate limiting for APIs

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9. Common Practical Interview Questions

Q66. How do you connect Node.js with MongoDB?

Answer:

Using Mongoose:

```js
const mongoose = require("mongoose");


mongoose.connect(process.env.MONGO_URI)
.then(() => console.log("MongoDB connected"))
.catch((error) => console.log(error));
```

Q67. How do you create an Express server?

Answer:

```JS
const express = require("express");
const app = express();


app.use(express.json());

app.get("/", (req, res) => {
res.send("API is running");
});

app.listen(5000, () => {
console.log("Server running on port 5000");
});
```

Q68. How do you create a Mongoose schema?

Answer:

```JS
const mongoose = require("mongoose");


const userSchema = new mongoose.Schema({
name: {
type: String,
required: true
},
email: {
type: String,
required: true,
unique: true
}
});

module.exports = mongoose.model("User", userSchema);
```

Q69. How do you fetch data in React from backend?

Answer:

```jsx
import { useEffect, useState } from "react";
import axios from "axios";


function Users() {
const [users, setUsers] = useState([]);

useEffect(() => {
async function fetchUsers() {
const res = await axios.get("http://localhost:5000/api/users");
setUsers(res.data);
}

fetchUsers();
}, []);

return (
<div>
{users.map(user => (
<p key={user._id}>{user.name}</p>
))}
</div>
);
}
```

Q70. How do you handle form input in React?

Answer:

```jsx
const [email, setEmail] = useState("");


<input
type="email"
value={email}
onChange={(e) => setEmail(e.target.value)}
/>
```

This is called a controlled component because React controls the input value.

Q71. How do you send form data from React to Express?

Answer:

```jsx
const submitHandler = async (e) => {
e.preventDefault();


await  axios.post("http://localhost:5000/api/users",  {
name,
email
});
};
```

The backend receives this data using:

```js
app.post("/api/users", (req, res) => {
console.log(req.body);
});
```

Q72. How do you update data in MongoDB using Mongoose?

Answer:

```js
const updatedUser = await User.findByIdAndUpdate(
req.params.id,
req.body,

{ new: true }
);
```

The `{ new: true }` option returns the updated document.

Q73. How do you delete data in MongoDB using Mongoose?

Answer:

```JS
await User.findByIdAndDelete(req.params.id);
```

This deletes a document by its ID.

Q74. What is the difference between PUT and PATCH?

Answer:

PUT	PATCH
Updates entire resource	Updates partial resource
Replaces old data	Modifies selected fields
Used for full update	Used for partial update

Q75. What is API testing?

Answer:

API testing means checking whether backend APIs work correctly.

Common tools:

- Postman
- Thunder Client
- Insomnia

Example API test:

```HTTP
GET http://localhost:5000/api/users
```

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10. MERN Project-Based Interview Questions

Q76. Explain your MERN project architecture.

Answer:

A typical MERN project has two main folders:

```
TEXT
project/
client/
React frontend
server/
Node and Express backend
```

The frontend handles UI and user interaction. The backend handles APIs, authentication, database operations, and business logic. MongoDB stores the application data.

Q77. How do you manage authentication in your MERN project?

Answer:

Authentication is usually handled using JWT and bcrypt.

Steps:

1. User registers with name, email, and password.
2. Password is hashed using bcrypt.
3. User data is saved in MongoDB.
4. During login, password is compared with hashed password.
5. If valid, JWT token is generated.
6. Token is sent to frontend.
7. Frontend stores token and sends it with protected API requests.

Q78. How do you protect API routes?

Answer:

API routes are protected using authentication middleware. The middleware verifies the JWT token. If the token is valid, the user can access the route. If not, the server returns an unauthorized response.

Q79. How do you handle errors in a MERN application?

Answer:

Errors can be handled using:

- `try-catch` blocks
- Express error-handling middleware
- Proper HTTP status codes
- Frontend error messages
- Logging errors for debugging

Example:

```
JS
try{
const
      users = await User.find();
res.json(users);
} catch (error) {
res.status(500).json({ message: error.message });
}
```

Q80. How do you improve performance in a MERN application?

Answer:

Performance can be improved by:

- Using MongoDB indexes
- Implementing pagination
- Optimizing React components
- Avoiding unnecessary re-renders
- Using lazy loading
- Compressing API responses
- Caching frequently used data
- Reducing large bundle sizes
- Using CDN for static assets

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11. HR and Scenario-Based MERN Questions

Q81. Why did you choose MERN Stack?

Answer:

I chose MERN Stack because it allows full-stack development using JavaScript. React makes frontend development interactive, Node and Express make backend development efficient, and MongoDB

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provides a flexible database. MERN is also widely used in modern web development and has strong community support.

Q82. What was the biggest challenge in your MERN project?

Answer:

A good sample answer:

“One challenge I faced was implementing authentication and protected routes. I solved it by using JWT for login sessions, bcrypt for password hashing, and middleware to protect backend routes. On the frontend, I used conditional routing to prevent unauthorized users from accessing private pages.”

Q83. How do you debug a MERN application?

Answer:

I debug a MERN application by:

- Checking browser console errors
- Checking backend terminal logs
- Testing APIs using Postman
- Verifying database records in MongoDB Compass or Atlas
- Using `console.log` carefully
- Checking network requests in browser developer tools
- Reading error messages properly

Q84. What will you do if an API is not working?

Answer:

I would check:

1. Is the backend server running?
2. Is the API URL correct?
3. Is the HTTP method correct?
4. Is the request body correct?
5. Are headers required?
6. Is CORS configured?

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7. Is the database connected?
8. Is there an error in backend logs?

Q85. How do you handle a slow MERN application?

Answer:

I would:

- Check API response time
- Add database indexes
- Use pagination
- Optimize React rendering
- Reduce unnecessary API calls
- Compress images
- Use lazy loading
- Check server performance
- Analyze network requests

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12. Quick Revision Table

Topic	Important Points
MongoDB	NoSQL database, collections, documents, indexes
Express.js	Routing, middleware, APIs, error handling
React.js	Components, state, props, hooks, routing
Node.js	Runtime, event loop, npm, async programming
Authentication	JWT, bcrypt, protected routes
API	REST, CRUD, HTTP methods
Deployment	Vercel, Netlify, Render, MongoDB Atlas
Secur ity	Validation, hashing, CORS, environment variables

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13. Most Asked MERN Interview Questions

1. What is MERNStack?
2. Why use MongoDB in MERN?
3. What is Express.js?
4. What is middleware?
5. What is React?
6. What is JSX?
7. What is the difference between state and props?
8. What is useEffect?
9. What is Node.js?
10. What is npm?
11. What is REST API?
12. What is CRUD?
13. What is JWT?
14. What is bcrypt?
15. How do you connect MongoDB with Node.js?
16. How does React communicate with backend?
17. What is CORS?
18. What is protected routing?
19. How do you deploy a MERN app?
20. How do you secure a MERN application?

14. Final Interview Preparation Tips

- Prepare atleast one MERN project explanationproperly.
- Practice CRUD APIs using Express and MongoDB.
- Understand React hooks clearly.
- Learn JWT authentication flow.
- Be ready to explain your folder structure.
- Practice API testing using Postman.
- Revise HTTP methods and status codes.
- Learn how frontend and backend communicate.

- Prepare real examples from your project.
- Do not memorize only definitions; understand the flow.

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15. Sample Self-Introduction for MERN Interview

Answer:

“My name is Dipanshu. I am interested in full-stack web development, especially using the MERN Stack. I have knowledge of MongoDB, Express.js, React.js, and Node.js. I can build frontend interfaces using React, create REST APIs using Express and Node.js, and store data in MongoDB. I have also worked with concepts like authentication, CRUD operations, routing, API integration, and deployment. I am continuously improving my problem-solving and development skills.”

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16. Conclusion

MERN Stack interviews usually test both theory and practical understanding. Focus on how the four technologies work together: React handles the frontend, Express and Node handle the backend, and MongoDB stores the data. If you can clearly explain project flow, authentication, API communication, and CRUD operations, you will be well prepared for most MERN Stack interviews.